

# Julius Alexander Zeiss, M.Sc. Physics

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## Profile

- 📌 Interested in the Mathematical-theoretical Foundations of Quantum Information
- 📌 International Collaborations tackling Problems from Quantum Causality
- 📌 Teaching Experience in Mathematics, Theoretical and Applied Physics

## Education

- 10/2019 – 08/2022 📌 **M.Sc. Physics, University of Cologne & University of Bonn, Germany**  
Focus: Theoretical Particle Physics, General Relativity, Mathematical Physics  
Thesis: *SDP Relaxations for Quantum Inflation*  
Advisor: Prof. Dr. David Gross, Thesis Grade: 1.0 (very good)  
Overall Grade: 1.4 (very good)
- 10/2018 – 08/2019 📌 **M.Sc. Physics, University of Münster, Germany**  
Focus: Theoretical Nuclear & Particle Physics, Theoretical Non-linear Physics, Quantum Information Theory, Biophysics
- 10/2015 – 09/2018 📌 **B.Sc. Physics, University of Münster, Germany**  
Thesis: *Numerical Treatment of the Manning Potential for the Description of Ammonia*  
Advisor: Prof. Dr. Gernot Münster, Thesis Grade: 1.0 (very good)  
Overall Grade: 1.8 (good)
- 04/2013 – 09/2015 📌 **B.Sc. Business Administration, University of Münster, Germany**  
(discontinued, transferred to Physics)
- 09/2010 – 05/2012 📌 **International Baccalaureate, King William's College, Isle of Man**  
Overall Grade: 41/45 Points (very good, 1.1 German Abitur)

## Teaching

- 04/2021 – 04/2022 📌 **University of Cologne, Germany**  
Tutorials on Linear Algebra & Analysis, Proofs in Mathematics, Theoretical Physics
- 04/2019 – 09/2019 📌 **University of Münster, Germany**  
Tutorials on Applied Physics

## Research Experience

- 02/2023 📌 **Perimeter Institute for Theoretical Physics, Waterloo, Canada**  
Research in Quantum Causality, Talk: *Infinite Dimensional Optimisation Problems in Quantum Information – An Operator Algebra Approach to the NPA Hierarchy*
- Currently 📌 **Collaboration with Laurens Ligthart, Prof. Dr. David Gross, University of Cologne, Germany**  
Completeness Proofs of Relaxation Hierarchies from Operator Algebra Arguments for Quantum Causality
- 05/2022 📌 **Mathematics Department, University of Klagenfurt, Austria**  
Efficiently Solving Large-scale Semidefinite Programming Problems from Quantum Causality, Talk: *SDP Relaxations for Quantum Inflation*

## Research Experience (continued)

- 01/2022 – 03/2022     **Collaboration with Dr. Elie Wolfe, Perimeter Institute of Theoretical Physics, Waterloo, Canada & Dr. Alejandro Pozas-Kerstjens, Complutense University of Madrid, Spain**  
Convex Optimisation of Non-commutative Polynomial Optimisation Problems

## Workshops

- 09/2022     **Quantum Correlations of Nature, University of Siegen, Germany**  
Research Discussions on Problems from the Field of Quantum Correlations
- 08/2022     **Quantum Computing, Bad Honnef Physics School, Germany**  
Introduction to the Theoretical Description of Quantum Computing
- 08/2019     **Programming in Artificial Intelligence & Machine Learning Intensive Week, University of Münster, Germany**  
Python Programming Applications for Artificial Intelligence and Machine Learning

## Skills

- Languages     German (native), English (bilingual)
- Coding     Python, Fortran, Mathematica, PHP, JavaScript, HTML, SQL, VBA

## References

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| <b>Prof. Dr. David Gross</b><br>Master Thesis Advisor<br>University of Cologne, Germany<br>david.gross@thp.uni-koeln.de | <b>Prof. Dr. Angelika Wiegele</b><br>Research Collaborator<br>University of Klagenfurt, Austria<br>angelika.wiegele@aau.at |
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## Outreach Articles

- 09/2022     **DPG Physics Journal Article**  
Review Article on the Quantum Computing Bad Honnef Physics School

## Outside of Academia – Employment

- 02/2019 – 09/2019     **Bar Levin & Co. KG, Münster, Germany**  
Bartender
- 06/2018 – 01/2019     **Freisfeld GmbH & Co. KG, Münster, Germany**  
Sales Assistant
- Summer 2015, Autumn 2014     **Fielmann AG, Hamburg, Germany**  
DCF Modelling with Monte-Carlo Simulations, VBA & SQL Solutions
- 01/2013 – 02/2013     **HMT, Hamburg, Germany**  
Mathematical Modelling using Monte-Carlo Simulations
- 08/2011     **AKO Capital, London, England**  
Internship Financial Research using Mathematical Modelling